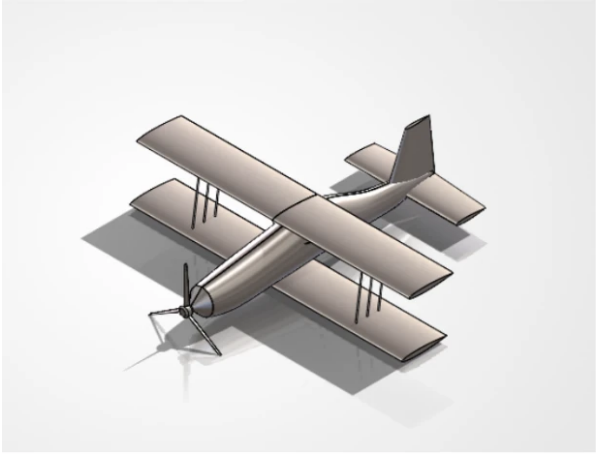
Aerobatic Biplane

Fall 2026 · In development · CAD development and analysis adaptation

I am developing a two-person aerobatic biplane concept through SolidWorks modeling and a modular MATLAB analysis. The project connects geometry with lift, drag, fuel use, power requirements and longitudinal trim so I can examine how a design choice affects the rest of the aircraft.



Actual SolidWorks views of my aircraft model in development. Wing, fuselage and tail geometry are being refined alongside the analysis.

Geometry and shared inputs

The SolidWorks concept is being developed alongside a modular MATLAB analysis adapted from an earlier aircraft workflow. Shared geometry, weights and mission assumptions connect the modules. The reference configuration uses 20 ft span, 127.5 ft² total wing area, 17 ft 9 in overall length and 260 hp sizing power, based on the Aviat Pitts S-2C. The project geometry is not an exact production-aircraft model.

Implemented methods

The ten MATLAB functions and one representative NACA 0012 data file cover wing and tail geometry, analytical section data, finite-wing lift, component drag, phase fuel and weight bookkeeping, power requirements, preliminary takeoff and landing estimates, and longitudinal trim. Internal consistency checks are included. Their presence does not prove that this source review executed or passed them.

Status and next steps

The output is developing CAD geometry and a preliminary analysis workflow. Next steps include reconciling CAD and analysis geometry and replacing estimated mass stations and areas with CAD properties. The section polar is analytical teaching data; no wind-tunnel or flight-test validation, structural substantiation or dynamic flight model is claimed.

Evidence basis: Student-supplied SolidWorks screenshot and bws013 MATLAB source archive; Aviat Pitts S-2C reference specifications: aviataircraft.com/fusion_element/pitts-specs/.